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WORK EXPERIENCE

Machine Learning & Fullstack Engineer (Intern), KLASS Engineering and Solutions Sep 2025 – Aug 2026

- Reproduced and benchmarked state-of-the-art vision and audio AI models (NSG-VD, D3, ReStraV, Speech-Forensics) to evaluate real-time transparency and explainability in threat detection against proprietary pipelines.
- Integrated advanced optical flow (RAFT), segmentation (SAM 2), and point-tracking algorithms (CoTracker) to analyze physical motion derivatives in video sequences for mission-critical data.
- Engineered a comprehensive data generation pipeline linking video extension models (Wan2.2), voice cloning (Chatterbox), and lip-synchronization (LatentSync) to curate a balanced 1,200-video dataset for benchmarking.
- Managed GPU-accelerated experimentation on shared SLURM clusters (NVIDIA H200 nodes) via Docker containers, and deployed fullstack architectures (React, Zustand, FastAPI, WebSockets) for multi-model AI endpoints.
- Designed scalable dataset curation workflows with custom validation splits and lightweight sanity-check protocols before full-scale benchmarking.

Volunteer / Full Time Research Assistant, Singapore University of Technology & Design (SUTD) - ROAR Lab Aug 2022 – May 2024

- Developed perception software and processed 3D LiDAR data for autonomous mobile robots, designing architectures integrating Embedded C++, ROS, and sensor fusion.
- Built a specialized predictive safety model to detect a stair-climbing robot's fall risk, deploying Long Short-Term Memory networks (LSTMs) to improve physical safety metrics by 50%.
- Led the design and development of an intelligent lighting control system for an autonomous vehicle, integrating it seamlessly with the main navigation system via RTOS and APIs.

Student Research Engineer, Singapore Institute of Technology (SIT) Aug 2024 – May 2025

- Designed and implemented an experimental temperature chamber to simulate human body heat signatures and extreme physical environments for environmental robotics testing.
- Engineered a closed-loop, thermistor-based control system to regulate precise environmental variables, ensuring high-fidelity data collection.
- Established repeatable testing and data collection protocols for faculty-led embodied AI and robotics hardware projects.

PROJECTS

1. AI-Generated Video & Audio Forensics Benchmarking Sep 2025 – Aug 2026

- Reproduced and benchmarked state-of-the-art vision and audio AI models (NSG-VD, D3, ReStraV, Speech-Forensics) to evaluate real-time transparency and explainability in threat detection against proprietary pipelines.
- Integrated advanced optical flow (RAFT), segmentation (SAM 2), and point-tracking algorithms (CoTracker) to analyze physical motion derivatives in video sequences for mission-critical data.

2. Synthetic Multimodal Deepfake Dataset Pipeline Sep 2025 – Aug 2026

- Engineered a comprehensive data generation pipeline linking video extension models (Wan2.2), voice cloning (Chatterbox), and lip-synchronization (LatentSync) to curate a balanced 1,200-video dataset for benchmarking.

3. Multi-Model AI Endpoint Platform Sep 2025 – Aug 2026

- Managed GPU-accelerated experimentation on shared SLURM clusters (NVIDIA H200 nodes) via Docker containers, and deployed fullstack architectures (React, Zustand, FastAPI, WebSockets) for multi-model AI endpoints.

4. Stair-Climbing Robot Fall-Risk Prediction (LSTM) Aug 2022 – May 2024

- Built a specialized predictive safety model to detect a stair-climbing robot's fall risk, deploying Long Short-Term Memory networks (LSTMs) to improve physical safety metrics by 50%.

SKILLS / COMPETENCIES

AI, Vision & Audio Analytics

1. Deep Learning
2. Computer Vision
3. Deepfake Detection
4. Generative AI
5. OpenCV
6. Optical Flow (RAFT)
7. Segmentation (SAM 2)
8. Point-Tracking (CoTracker)

Machine Learning Frameworks

1. PyTorch
2. TensorFlow
3. Keras
4. Hugging Face Transformers
5. CNNs
6. DNNs
7. LSTMs
8. YOLO
9. Vision Transformers
10. Reinforcement Learning
11. Model Benchmarking
12. Zero-shot Learning

Software & Fullstack

1. Python (NumPy, Pandas)
2. C++17
3. C#
4. Java
5. Embedded C
6. TypeScript
7. React
8. Zustand
9. FastAPI
10. OpenAPI
11. WebSockets
12. Git
13. Linux/Ubuntu
14. WSL

Infrastructure & Cloud Tech

1. SLURM
2. GPU Computing (CUDA)
3. NVIDIA HPC Clusters (H200 nodes)
4. Docker
5. Dev Containers
6. ONNX Runtime
7. uv
8. IoT Integrations
9. Agile/Scrum Methodologies

EDUCATION

DigiPen Institute of Technology & Singapore Institute of Technology Aug 2023 – May 2027

- B.Eng. Mechatronics Systems

- Relevant Coursework: Mechanics, Control Systems, Robotics, Embedded Systems, Machine Learning, Systems and Software Engineering

Ngee Ann Polytechnic Sep 2022 – Sep 2023

- Specialist Diploma in Robotics Engineering (SGInnovate PowerX)
- Relevant Coursework: Autonomous Robot Navigation, Machine Learning/Deep Learning for Robotics, Computer Vision

Singapore Polytechnic Apr 2017 – Sep 2020

- Diploma in Mechatronics and Robotics
- Internship grade of B+; coursework in Microcontroller Applications, PLCs, and Robotics Integration

AWARDS / ACHIEVEMENTS / CERTIFICATES

Award / Certificate	Date
MOE (Ministry of Education) Instructor Certification (Achieved at age 18)	Valid Oct 2018 – Oct 2026
Google Crash Course on Python	
Google The Bits and Bytes of Computer Networking	
Coursera Machine Learning	